Submission Cover Letter & Author Declaration

Target Journal: *npj Digital Medicine* (Nature Portfolio)
Article Type: Article (Original Research / Evidence Audit)
Manuscript Title: Claimed quantum benefits in clinical machine learning do not survive symmetric evaluation
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To the Editors of *npj Digital Medicine*:

I am pleased to submit our original research article, **"Claimed quantum benefits in clinical machine learning do not survive symmetric evaluation,"** for consideration for publication in *npj Digital Medicine*.

Scientific Context and Rationale for *npj Digital Medicine*

Clinical artificial intelligence is currently experiencing rapid methodological diversification. Across digital health, novel model architectures—from foundation model adapters and attention-gated heads to hybrid classical-quantum circuits—are routinely reported to deliver superior predictive performance. However, evaluating whether an architectural component offers genuine inductive efficiency versus apparent superiority driven by asymmetric care remains a pervasive challenge. When a novel model is aggressively tuned and calibrated while standard classical baselines are implemented with default settings, evaluated under global pre-split dimensionality reduction, or assessed without paired inferential tests, biomedical researchers and procurement committees can be severely misled.

This paper addresses this challenge by examining quantum machine learning (QML) in digital health—a high-profile paradigm where empirical claims of superiority have proliferated despite theoretical limits at low qubit widths (4 to 12 qubits). Following a recent systematic review in *npj Digital Medicine* (Gupta et al., 2025) which synthesized published metrics and observed negligible differences, our study goes substantially deeper: we directly audit the raw code, data partitioning, hyperparameter spaces, and binary prediction arrays to investigate *why* these discrepancies occur.

Central Contributions and Key Empirical Findings

We formulate an operational, reproducible **seven-rule symmetric evaluation protocol** and an actionable **17-item audit checklist** designed to eliminate methodological asymmetries in clinical AI evaluations. We apply this protocol across both an internal clinical decision-support prototype and an independent field audit of the published literature:

1. **Internal Pre-Registered Confirmatory Test ($N = 56,534$ Patients):**
2. Using an untouched test cohort of 56,534 independent human patients from CheXpert (one frontal radiograph per patient; verified zero patient overlap with development data), we evaluated a variational quantum circuit against matched classical baselines for medical triage.

3. Under default matched optimization (Analysis C1), the quantum circuit was materially worse on negative log-likelihood ($\Delta\text{NLL} = +0.0205$, 95% CI 0.0186 to 0.0224; $t = 21.034, p = 7.65 \times 10^{-98}$; positive indicates worse likelihood).
4. Removing entanglement improved likelihood across initialization seeds ($p = 0.0013$), an effect traced to inductive bias mismatch on an additive clinical triage rule.

5. When granted symmetric hyperparameter search budgets over 36 configurations (Analysis C2), the optimal quantum configuration achieved a small, statistically detectable likelihood advantage over linear Product-of-Experts ($\Delta\text{NLL} = -0.0026$ nats, 95% CI -0.00359 to -0.00160, $p = 3.07 \times 10^{-7}$) and nonlinear Gated Heads ($\Delta\text{NLL} = -0.0012$ nats, 95% CI -0.00192 to -0.00041, $p = 0.0025$). Crucially, this observed difference fell entirely within our pre-registered operational equivalence margin (± 0.010 nats; Two One-Sided Test $p_{\text{TOST}} < 10^{-6}$), confirming practical statistical equivalence rather than transformative clinical utility.

6. **Systematic Field Audit of Published Literature (5 Reimplemented Studies, 6 Tasks):**

7. From 231 screened PubMed records, 8 publications met eligibility criteria, of which 5 publications spanning 6 clinical prediction tasks were reconstructible and reimplemented with full code (3 were non-reconstructible due to unreleased code, proprietary radiomics, or restricted data).

8. Applying our prespecified survival criterion requiring statistically significant superiority under leak-free nested cross-validation, **zero of the six published advantage claims survived symmetric evaluation**:

    - **1 Claim Reversed:** Chronic Kidney Disease (Parthasarathi et al., 2025; classical models won by 9.65 percentage points, $p = 0.0009$).
    - **4 Claims Statistically Inconclusive:** Parkinson's disease voice ($p = 0.98$), Pima diabetes ($p = 0.067$), Glioma grading ($p = 0.782$ on TCGA; $p = 0.191$ on BraTS 2020), and Heart disease risk ($p = 0.310$ uncorrected; Nadeau–Bengio corrected $p = 0.897$, 95% CI -18.45% to $+16.25\%$).
    - **1 Claim Collapsed to Chance Level:** Schizophrenia EEG (Aksoy & Cattan, 2024; the reported 100% accuracy reproduced under random epoch splitting, but collapsed to 52.1% chance level when subject-level temporal partition leakage was eliminated).

9. **Mechanistic and Theoretical Clarification:**

10. Through exact Fourier expansion (M1), classical trigonometric surrogates (M2; outperforming the quantum circuit by $\Delta\text{NLL} = 0.0238$ to 0.0267, $p < 10^{-4}$), and geometric kernel distance bounds ($g(K_{\text{RBF}} \parallel K_Q) = 1.2712 \ll 44.72$), we mathematically isolate why shallow quantum circuits provide no inductive advantage over classical models in these tabular and imaging regimes.

Open Science, Reproducibility, and Reporting Standards

- **Pre-Registration:** The confirmatory protocol, primary endpoints, equivalence margin (± 0.010 nats), and decision rules were pre-registered and deposited in `docs/PREREGISTRATION_v6_2026-10-03.md` (Git commit `61e777b`) on October 3, 2026, eight days prior to test data scoring on October 11, 2026.
- **Reporting Guidelines:** The manuscript is fully compliant with **TRIPOD+AI** (line-by-line item mapping provided in Supplementary Note 10) and **PRISMA 2020** (Supplementary Note 6).

- **One-Command Reproduction:** All experimental code, simulation harnesses, and binary prediction artifacts are openly available at `https://github.com/Sivaguru-2008/aura`. The entire Level B numerical verification suite (recomputing all reported metrics, forest plots, and p-values from raw prediction checkpoints) executes in under 12 seconds via `python reproduce_paper.py`.

Author Declaration & Mandatory Statements

1. **Originality and Prior Publication:** The manuscript is original work, has not been published previously, and is not currently under consideration for publication elsewhere.

2. **Ethical Compliance and Data Governance:** This study involves secondary analysis of established, fully de-identified clinical and biomedical databases:

3. MIMIC-CXR-JPG (v2.0.0) under PhysioNet Credentialed Health Data License 1.5.0.
4. CheXpert-v1.0-Small under the Stanford University Research Data Agreement.
5. BraTS 2020 via the Synapse repository.
6. UCI Machine Learning Repository cohorts (CKD, Parkinson's Voice, Pima Diabetes, Heart Disease).
7. MSU Adolescent Schizophrenia EEG database.

8. The Cancer Genome Atlas (TCGA) Glioma project via NIH Genomic Data Commons. All patient records were de-identified in source repositories, and no direct human participant intervention was performed.

9. **Competing Interests:** The author declares no competing financial or non-financial interests.

10. **Data Availability:** All public databases are accessible via their respective credentialed repositories. All derived model parameter checkpoints, per-patient prediction arrays, and numerical summary ledgers are deposited in the open GitHub repository.

11. **Code Availability:** All execution code, custom PyTorch quantum statevector engines, nested cross-validation harnesses, and reproduction scripts are released openly under an MIT license at `https://github.com/Sivaguru-2008/aura`.

Suggested Reviewers

We suggest the following independent experts in clinical machine learning validation, biostatistics, reproducibility, and quantum computing in healthcare:

1. **Prof. Ben Glocker, PhD**
   Affiliation: Department of Computing, Imperial College London, UK
   Expertise: Machine learning for medical image analysis, algorithmic fairness, validation pitfalls, and reproducibility in digital health.
   Rationale: Authored leading critiques on evaluation pitfalls in medical AI (*Lancet Digital Health*, 2022).

2. **Prof. Gary S. Collins, PhD**
   Affiliation: Centre for Statistics in Medicine, University of Oxford, UK
   Expertise: Medical statistics, clinical prediction model validation, and lead author of the TRIPOD and TRIPOD+AI reporting guidelines.

Rationale: World authority on methodological rigor, data leakage prevention, and reporting integrity for clinical AI.

3. **Prof. Maria Schuld, PhD**
   Affiliation: Department of Physics, University of KwaZulu-Natal / Xanadu Quantum Technologies
   Expertise: Quantum machine learning theory, kernel methods, Fourier analysis of variational circuits, and inductive bias.
   Rationale: Developed foundational theorems on data encoding and Fourier expressivity in variational quantum circuits (*Phys. Rev. A*, 2021).

4. **Prof. Hsin-Yuan (Robert) Huang, PhD**
   Affiliation: Institute for Quantum Information and Matter, California Institute of Technology (Caltech), USA
   Expertise: Quantum learning theory, sample complexity, and geometric difference bounds between classical and quantum kernels.
   Rationale: Developed the geometric kernel distance framework ($g(K_C \parallel K_Q)$) evaluated in our mechanistic analysis (*Nature Communications*, 2021).

5. **Prof. Judy Wawira Gichoya, MD, MS**
   Affiliation: Department of Radiology and Imaging Sciences, Emory University School of Medicine, USA
   Expertise: Clinical chest radiography AI, dataset bias, and validation of medical imaging decision support.
   Rationale: Expert in translational evaluation and validation of deep learning models on CheXpert and MIMIC-CXR.

Thank you for your time, consideration, and editorial guidance. We look forward to your response.

Sincerely,

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